Unit 1 – Using Objects and Methods Name: KEY
String and Math Methods Stations Activity

Directions: Code the solutions to the problems on the task cards around the room. You do not have to write a class with a main method, just the line or lines of code are fine. A timer will be set for each station. Do not move on until the timer tells you to rotate. After all 10 station solutions are written, answer the questions that follow. Turn this in at the end of class.

Station 1 Scanner keyboard = new Scanner(System.in); System.out.print("Word: "); String word = keyboard.next(); System.out.println("Characters: " + word.length());	Station 2 String word = "JavaProgramming"; System.out.println(word.substring(4,11));
Station 3 String word = "Hello, world!"; System.out.println(word.indexOf("w"));	Station 4 Scanner keyboard = new Scanner(System.in); System.out.print("Email: "); String word = keyboard.next(); System.out.println("Username: " + word.substring(0, word.indexOf("@")));
Station 5 double result = Math.pow(3, 2) + Math.sqrt(25);	Station 6 double result = Math.abs(-15-10);
Station 7 double result = 1000*Math.pow(1.05,2);	Station 8 int result = "apple".compareTo("banana");

Station 9
 	
int roll = (int)(Math.random() * 20) + 1;
int dex = 5;
System.out.println("Initiative: " + (roll + dex));
	Station 10
	
	String word = "Code:Winner123";
	System.out.println(word.substring(5, 11));

Wrap-Up Questions *(sample student responses)*

1) How do `.substring()` and `.indexOf()` work together to extract parts of a string?

If we have `String sentence = "I love programming!"`, we can find the position of "love" using `sentence.indexOf("love")`, then use `substring()` to extract "love programming!" by calling `sentence.substring(sentence.indexOf("love"))`. This is useful when we need to get parts of a String extracted but we don't know where it starts.

2) Why do we need `Math.abs()` in real-world applications?

One example is in navigation. If we're calculating the distance between two points, we don't want negative distances, so we take the absolute value of the difference in coordinates.

3) Describe a real-world problem where you might need to combine multiple string methods.

If we have a USPS barcode and needed only a specific part extracted for their inventory, we could use a combination of String methods like in question 1 to get the code part we need.